

Pengxiang (Eric) Wang

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EDUCATION

University of Toronto

Sep 2023 – Apr 2028 (Expected)

B.Sc. Computer Science (Specialist) GPA: 3.78 / 4.00

Coursework: CSC413 Neural Networks & Deep Learning, CSC415 Reinforcement Learning, CSC384 Intro to Artificial Intelligence, CSC311 Intro to Machine Learning, CSC373 Algorithm Design & Analysis, CSC363 Computational Complexity & Computability, CSC369 Operating Systems

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL
AI / ML: PyTorch, HuggingFace Transformers, model distillation, RAG, ReAct agents
Experiment: ablation studies, human evaluation (MOS), Whisper WER, statistical analysis
LLM Infra: Spring AI, LangChain4j, PgVector, Ollama, MCP, SSE streaming
Backend: Spring Boot, PostgreSQL, Redis, Docker
Frontend: React, Vue

PROJECTS

Circuit Agent — Full-Stack LLM Agent Platform

Spring AI 1.0 · Java 21 · PgVector · SSE · MCP · Docker | [GitHub](#)

- Built a dual-application platform (domain RAG chatbot + autonomous ReAct agent with 7 pluggable tools) to explore how modern LLM agent patterns work under the Java/Spring ecosystem.
- Reduced perceived response latency by switching from buffered to SSE streaming, cutting time-to-first-token from $\sim 1.2s$ to $\sim 0.5s$ under a 20-step deterministic execution budget.
- Improved RAG retrieval relevance by adding a pre-retrieval query rewriting step and index-time keyword enrichment per chunk, lifting answer quality noticeably on domain-specific questions.
- Designed a 4-layer agent hierarchy (**BaseAgent** $\rightarrow$ **ReActAgent** $\rightarrow$ **ToolCallAgent** $\rightarrow$ **EkManus**) to enforce deterministic loop control and prevent uncontrolled recursive execution.
- Scaled concurrent agent sessions by routing ReAct loops through async execution on Java 21 virtual threads, keeping SSE streams non-blocking.

NeuTTS-Express — Expressive TTS via Synthetic Distillation

PyTorch · Knowledge Distillation · HuggingFace Transformers · Neural Audio Codec | [GitHub](#) | [Paper](#)

- Lightweight TTS models are fast but lack emotional expression; designed a synthetic data pipeline where a 3B teacher model generates emotion-labeled audio to train a 0.5B student, avoiding costly human-annotated datasets.
- Built an evaluation pipeline measuring Whisper WER, real-time factor, and batch stability; student model runs $2.16\times$ faster than real-time with $\sim 4\times$ lower WER variance (std 2.1% vs. 7.9%).
- Ran an 11-participant MOS study across 7 emotions; the student outperformed its $6\times$ larger teacher on naturalness (3.99 vs. 3.17) and emotion match (3.91 vs. 2.82).

Shield-Guided Policy Stabilization for Safe Reinforcement Learning

JAX · DreamerV3 · AMBS · Python | [GitHub](#) | [Paper](#)

- In shielded RL, the task policy never internalizes safety rules and oscillates across safety boundaries during training; proposed two complementary fixes: a threshold-based cost penalty and KL-regularized actor updates.
- Ran an 8-group ablation (baseline / cost-penalty / KL / full) across 3 safety labels in Atari Seaquest; the full method cut total safety violations from 272 to 112 over 45k steps while keeping task reward competitive.

Cloud Gallery — Collaborative Cloud Image Platform

Spring Boot 2.7 · Java 11 · Vue 3 · WebSocket · Disruptor · Cloudflare R2 · Redis | [GitHub](#)

- Eliminated cross-tenant data leakage across concurrent sessions by isolating two independent Sa-Token **StpLogic** instances (system roles vs. space roles), enforced at every API boundary without coupling the two domains.
- Achieved deterministic conflict resolution for real-time collaborative edits by routing concurrent WebSocket writes through a Disruptor-backed event queue with a lock-editor protocol.
- Reduced image post-processing compute cost by configuring Cloudflare R2 CI rules to auto-trigger WebP compression ($\sim 30\%$ size reduction), thumbnail generation, and dominant-color extraction on every upload.
- Delivered color-based image search at **sub-millisecond latency**, no external embedding model, by normalizing dominant colors to HSL and ranking via Euclidean RGB distance in a SQL aggregation.